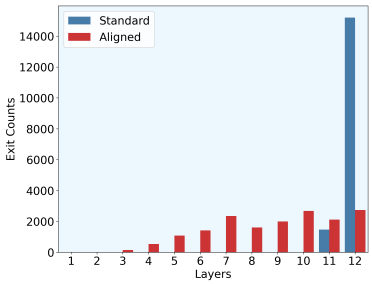
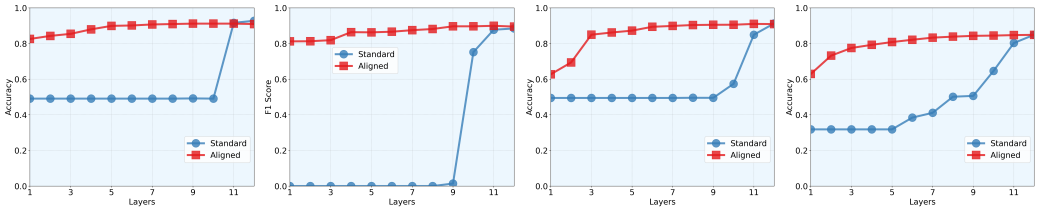
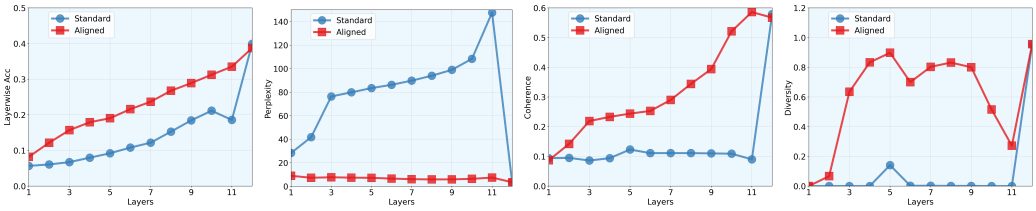


Figure 7: Number of samples exiting at each layer.



(a) SST-2 (b) MRPC (c) QNLI (d) MNLI

Figure 8: Layerwise accuracy for **AlignedBERT** and BERT and on SST-2, MRPC, QNLI and MNLI datasets of the GLUE benchmark. (See more results on RTE and QQP in Figure 21).



(a) Prediction Accuracy (b) Perplexity (c) Coherence (d) Diversity

Figure 9: Evaluation of **AlignedGPT** and GPT2 on Wikitext-103 dataset in terms of (a) prediction accuracy, (b) perplexity, (c) coherence, and (d) diversity. See Appendix C.2 for detailed definitions.

trained with aligned training achieves 77.96% accuracy with a $1.36\times$ speedup, which is comparable to those trained by multi-exit/classifiers with 78.32 % accuracy and $1.42\times$ speedup.

Effects on Transferability Concerns arise about whether aligning shallow layer features with deep layer features could diminish the transferability of shallow layers. To answer this question, we conduct experiments on (1) distribution shift and (2) task transfer. The results show that aligned training improves layer-wise accuracy for both pre-trained and downstream datasets while maintaining transferability. This indicates that the trained model can be effectively transferred without losing the universal patterns learned in shallow layers. Further details are provided in Appendix B.4.

3.2 Applications on Language Models

We extend our aligned training approach to NLP tasks, demonstrating its effectiveness in fine-tuning Language Models. For text classification tasks, we get **AlignedBERT** by finetuning a pretrained 12-layer $\text{BERT}_{\text{Base}}$ model [6] using aligned training method on GLUE benchmark [32] tasks. This includes single-sentence tasks-SST-2, similarity and paraphrasing tasks-MRPC and QQP, as well as natural language inference tasks-MNLI, QNLI and RTE. For comparison, we also finetune a $\text{BERT}_{\text{Base}}$ model using standard training. Then we evaluate the layerwise accuracy(or F-1 score) of both finetuned models using their last layer classifier. Figure 8 shows that AlignedBERT achieves better performance than the standard BERT across all layers.

For text generation task, we get **AlignedGPT** by finetuning a pretrained 12-layer GPT2 model [7] using aligned training method on Wikitext-103 dataset [33] . For comparison, we also finetune a GPT2 model using standard training. Then we evaluate the two finetuned models from two per-

spectives(following [59, 60]), (1) language modeling quality, which assesses the intrinsic quality of the model and is measured by prediction accuracy and perplexity, and (2) generation quality, which measures the quality of the text produced by the model using coherence and diversity. Coherence is a measurement of relevance between prefix text and generated text, while diversity considers the recurrence of generation at varying n-gram levels. All evaluation metrics mentioned above can be found in Appendix C.2. Results show that AlignedGPT outperforms the standard one in prediction accuracy and exhibits lower perplexity across intermediate layers(Figure 9(a,b)). Moreover, AlignedGPT can also generate text with higher coherence and diversity using shallower layers(Figure 9(c,d)), which improves the the inference efficiency. More experimental setups can be found in Appendix C.2.

4 Conclusion

In this paper, we demonstrate that a simple sample-wise cosine similarity metric effectively captures layer-wise representation similarity in transformer models, aligning with the more complex CKA metric. Our findings reveal that representations become more similar as layers get closer and show that increased representation similarity correlates with higher prediction accuracy, leading to saturation events where shallow layers can already make correct predictions. To enhance this, we proposed an aligned training method that improves shallow layer effectiveness, resulting in more early saturation events and much higher layer-wise accuracies. Remarkably, when served as multi-exit models with a common classifier, which to our knowledge is the first of its kind, they maintain the early exit capability and achieve performance on par with models that use multiple classifiers. Experiments on both vision and NLP tasks demonstrate the performance of the proposed aligned training.

Acknowledgement

We acknowledge support from NSF grants CCF-2240708, IIS-2312840, and IIS-2402952. We are grateful to Beidi Chen (Carnegie Mellon University) and Chong You (Google Research) for many valuable discussions and for helpful comments on the manuscript.

References

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[3] Jiahui Yu, Xin Li, Jing Yu Koh, Han Zhang, Ruoming Pang, James Qin, Alexander Ku, Yuanzhong Xu, Jason Baldridge, and Yonghui Wu. Vector-quantized image modeling with improved vqgan. *arXiv preprint arXiv:2110.04627*, 2021.

[4] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International conference on machine learning*, pages 8821–8831. Pmlr, 2021.

[5] Jiahui Yu, Yuanzhong Xu, Jing Yu Koh, Thang Luong, Gunjan Baid, Zirui Wang, Vijay Vasudevan, Alexander Ku, Yinfei Yang, Burcu Karagol Ayan, et al. Scaling autoregressive models for content-rich text-to-image generation. *arXiv preprint arXiv:2206.10789*, 2(3):5, 2022.

[6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

[7] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.

[8] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.

[9] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

[10] Vardan Papyan, XY Han, and David L Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.

[11] Cong Fang, Hangfeng He, Qi Long, and Weijie J Su. Exploring deep neural networks via layer-peeled model: Minority collapse in imbalanced training. *Proceedings of the National Academy of Sciences*, 118(43):e2103091118, 2021.

[12] Zhihui Zhu, Tianyu Ding, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and Qing Qu. A geometric analysis of neural collapse with unconstrained features. *Advances in Neural Information Processing Systems*, 34:29820–29834, 2021.

[13] Christos Thrampoulidis, Ganesh Ramachandra Kini, Vala Vakilian, and Tina Behnia. Imbalance trouble: Revisiting neural-collapse geometry. *Advances in Neural Information Processing Systems*, 35:27225–27238, 2022.

[14] Tom Tirer, Haoxiang Huang, and Jonathan Niles-Weed. Perturbation analysis of neural collapse. In *International Conference on Machine Learning*, pages 34301–34329. PMLR, 2023.

[15] Hangfeng He and Weijie J Su. A law of data separation in deep learning. *Proceedings of the National Academy of Sciences*, 120(36):e2221704120, 2023.

[16] Akshay Rangamani, Marius Lindegaard, Tomer Galanti, and Tomaso A Poggio. Feature learning in deep classifiers through intermediate neural collapse. In *International Conference on Machine Learning*, pages 28729–28745. PMLR, 2023.